Circles as mirrors: the circle kei, and where two mirrors agree

Notes for Carlo Perassi — companion to the reply Short

31 October 2026

Abstract

We look at the operation "reflect circle D in circle C" as a binary operation $C \star D$ on circle space $\mathbb{P}^3$. It satisfies the axioms of a *kei* (involutory quandle). It is closed on non-degenerate circles, isometric, and neither commutative nor associative. For two mirrors K, L we prove that the points where they *agree* ($K \star P = L \star P$) are the two crossings together with the two limiting points. That makes four points over $\mathbb{C}$, of which exactly two are real. So the pair Inigo Quilez calls "hidden" is half of a single object, and its visible partner is where the two mirrors agree. Every numerical claim was checked in `alt_verify.py` (log: `alt_verify_log.txt`).

1 Circle space

A generalised circle $a(x^2+y^2)+bx+cy+d=0$ is a point $[a:b:c:d] \in \mathbb{P}^3$; we allow complex entries. Let

$$\langle X, Y\rangle = X^{\top} G Y, \qquad G = \begin{pmatrix} 0 & 0 & 0 & -2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -2 & 0 & 0 & 0 \end{pmatrix}, \qquad Q(X) = \langle X, X\rangle = b^2 + c^2 - 4ad.$$

G has signature $(3,1)$. For a real circle with centre (x_0, y_0) and radius r we have $X = [1 : -2x_0 : -2y_0 : x_0^2 + y_0^2 - r^2]$ and $Q = 4r^2$. Lines have $a = 0$ and $Q > 0$. Points (circles of radius 0) have $Q = 0$. Real "imaginary circles" (radius$^2 < 0$) have $Q < 0$. Two circles are orthogonal iff $\langle X, Y\rangle = 0$, and a point P lies on C iff $\langle P, C\rangle = 0$.

2 The operation

Definition 1. For $Q(C) \neq 0$ put

$$C \star D \;=\; s_C(D) \;=\; D - \frac{2\langle D, C\rangle}{Q(C)}\, C.$$

For real C this is inversion in C (reflection if C is a line), applied to the circle D. Since s_C is linear and homogeneous of degree 0 in C, it is well defined on $\mathbb{P}^3$. Let $\mathcal{X} = \{[C] \in \mathbb{CP}^3 : Q(C) \neq 0\}$ and let $\mathcal{X}_{\mathbb{R}}$ be its real points.

Proposition 2 (Isometry and closure)**.** *s_C is the orthogonal reflection in the hyperplane $C^{\perp}$. Hence $\langle s_C X, s_C Y\rangle = \langle X, Y\rangle$ and $Q(s_C X) = Q(X)$. In particular $\star$ is closed on $\mathcal{X}$ and on $\mathcal{X}_{\mathbb{R}}$, and it preserves the sign of Q on real circles.*

Proof. s_C fixes $C^{\perp}$ pointwise and sends $C \mapsto -C$. In $\langle s_C X, s_C Y\rangle$ the two cross terms $\mp 4\langle X, C\rangle\langle Y, C\rangle / Q(C)$ cancel. □

Theorem 3 (The circle kei)**.** *For all $C, D, E \in \mathcal{X}$:*

$$\text{(K1) } C \star C = C, \qquad \text{(K2) } C \star (C \star D) = D, \qquad \text{(K3) } C \star (D \star E) = (C \star D) \star (C \star E).$$

So $(\mathcal{X}, \star)$ is a kei (involutory quandle), and so is $(\mathcal{X}_{\mathbb{R}}, \star)$.

Proof. (K1) $s_C(C) = C - 2C = -C$, which is the same point of $\mathbb{P}^3$. The axiom holds projectively, not on vectors. (K2) $s_C^2 = \mathrm{id}$, as for any reflection. (K3) For an isometry g we have $g\, s_D\, g^{-1} = s_{gD}$, because $s_{gD}(X) = X - 2\langle X, gD\rangle gD / Q(gD) = g\, s_D(g^{-1}X)$. Taking $g = s_C$ gives $s_C s_D s_C = s_{C \star D}$. Applying both sides to $C \star E = s_C E$ gives (K3). □

Thus $C \mapsto s_C$ embeds $\mathcal{X}$ into the conjugation quandle of reflections in $O(4)$; this is the standard way symmetric spaces give involutory quandles. The fixed set of s_C on $\mathbb{P}^3$ is $\{[C]\} \cup \mathbb{P}(C^{\perp})$, so $[C]$ is an isolated fixed point. Hence $\mathcal{X}_{\mathbb{R}}^{Q>0}$ also satisfies Loos's axioms for a symmetric space [3, 4]. Keis go back to Takasaki [1], quandles to Joyce [2]; see Stanovský [5] for the history, including Euclidean reflections as the model example.

Numbers (verified). The operation is not commutative and not associative. Take K the unit circle, $L : y = 1.7$, and M the circle with centre $(1.5, 0.8)$ and radius 1. Then:

- $K \star L$ is the circle with centre $(0, 0.2941)$ and radius $0.2941 = 1/3.4$.
- $L \star K$ is the circle with centre $(0, 3.4)$ and radius 1.
- $(K \star L) \star M$ has centre $(0.0862, 0.3232)$ and radius 0.0574.
- $K \star (L \star M)$ has centre $(0.1873, 0.3246)$ and radius 0.1248.

Over 4000 random triples (real circles, lines, imaginary circles, complex vectors), the largest errors were:

- (K1) $3 \cdot 10^{-16}$, (K2) $4 \cdot 10^{-12}$, (K3) $1.5 \cdot 10^{-11}$;
- Q-invariance $3 \cdot 10^{-11}$, $\langle , \rangle$-invariance $1 \cdot 10^{-11}$.

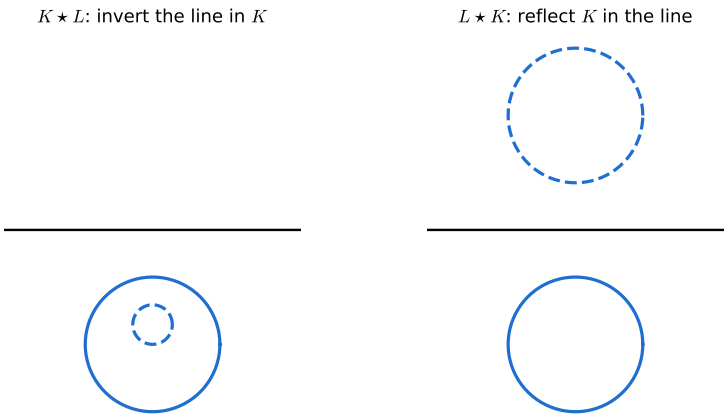


Figure 1: Order matters. Left: the line $y = 1.7$ inverted in the unit circle K becomes a circle through the centre. Right: K reflected in the line.

Remark 4 (Points cannot be mirrors)**.** If $Q(C) = 0$ the formula is undefined. Clearing the denominator gives $Q(C)D - 2\langle D, C\rangle C \sim C$, so every D goes to the point C. This matches the limit: inverting a fixed circle in a mirror of radius ε gives an image of radius $O(\varepsilon^2)$ that collapses onto the centre. Numerically, radius 0.01 gives an image of radius $7 \cdot 10^{-3}$, and radius 0.001 gives $7 \cdot 10^{-5}$. So points form a set that is *acted on* (s_C preserves $Q = 0$) but does not act. This is the domain caveat.

Lemma 5 (What a mirror fixes)**.** $\mathrm{Fix}(s_C) = \{[C]\} \cup \{X : \langle X, C\rangle = 0\}$*: the mirror itself, and every circle orthogonal to it. The fixed* points *($Q = 0$) are exactly the points lying on C.*

Proof. $s_C X = \lambda X$ forces either $X \parallel C$ or $\langle X, C\rangle = 0$. For a point P, $\langle P, C\rangle = 0$ means $P \in C$. □

3 Two mirrors: transport, common fixed circles, agreement

Transport. Since s_M is an isometry and preserves $Q = 0$, it carries the point pair $K \cap L$ to $(M \star K) \cap (M \star L)$. It preserves the invariant

$$I = \frac{\langle K, L\rangle^2}{Q(K)Q(L)},$$

so it sends a real pair to a real pair and a hidden pair to a hidden pair. In the checks, I was preserved for several values (0.36, 2.89, 0.0156, 4), and the pair was transported with error $3.5 \cdot 10^{-15}$.

Fix real mirrors $K \neq L$ with $Q > 0$, let $V = \mathrm{span}(K, L)$ and $U = V^{\perp}$. After normalising, the Gram matrix of V is $\begin{pmatrix} 1 & g \\ g & 1 \end{pmatrix}$ with $g^2 = I$.

- If $I < 1$ (the mirrors cut), V has signature $(2, 0)$, so U has signature $(1, 1)$.
- If $I > 1$ (disjoint or nested), V has signature $(1, 1)$, so U has signature $(2, 0)$.

The null lines of U are the crossings $K \cap L$, real or hidden. The null lines of V are the two *limiting points* of the pencil V.

Lemma 6 (Common fixed circles)**.** $\mathrm{Fix}(K) \cap \mathrm{Fix}(L) = \mathbb{P}(U)$ *(for $K \not\perp L$). If $I < 1$ this is the Apollonian family around the two real crossings. If $I > 1$ it is the pencil of real circles through the two limiting points.*

Proof. The first claim follows from Lemma 5. A pencil contains real point-circles if and only if it has signature $(1, 1)$. Its members then avoid those points. Signature $(2, 0)$ means its members all pass through the null points of $U^\perp = V$. Numerically, for $c = 1.7$, $\max |\langle u, P\rangle| = 1.4 \cdot 10^{-15}$. □

Theorem 7 (Where two mirrors agree — "always two points")**.** *Let*

$$\mathrm{Agree}(K, L) = \{P : \ Q(P) = 0, \ K \star P = L \star P \ \textit{in} \ \mathbb{P}^3\}.$$

Then:

1. $\mathrm{Agree}(K, L)$ *is the union of the null points of U (the crossings) and the null points of V (the limiting points). Both mirrors fix the crossings, and both mirrors swap the two limiting points.*
2. *Over $\mathbb{C}$ this gives four points (for $I \neq 0, 1$). Exactly two of them are real: the crossings if $I < 1$, the limiting points if $I > 1$. They merge into the tangency point if $I = 1$.*
3. $\mathrm{Agree}(K, L) \cap \hat{\mathbb{C}} = \mathrm{Fix}(s_L \circ s_K)$*, the fixed points of a Möbius transformation. So in the real plane two distinct mirrors agree at exactly two points, or at one if they are tangent (count ∞ for lines).*

Proof. $s_K P \sim s_L P$ is equivalent to $s_L s_K P \sim P$. Write $P = u + v$ with $u \in U$ and $v \in V$. The map $s_L s_K$ is the identity on U, and on V it is a rotation (if $I < 1$) or a boost (if $I > 1$) by twice the angle between K and L. Hence P is an eigenvector exactly when $P \in U$ or P lies on an eigenline of $s_L s_K|_V$. For $I \neq 0, 1$, the eigenlines of $s_L s_K|_V$ are the null lines of V. Intersecting with $Q = 0$ gives two points from U and two from V. Reality follows from the signatures above. On the limiting points $v_\pm$, s_K and s_L each act as the reflection of the hyperbolic plane V in a non-null line, so they swap the two null lines. For (3), the orientation-preserving composite $s_L s_K$ acts on $\hat{\mathbb{C}}$ as a Möbius map $z \mapsto (\alpha z + \beta)/(\gamma z + \delta)$. Its fixed points solve a quadratic. □

Our configuration. Take $K = \{|z| = 1\}$ and $L = \{y = c\}$. Then $s_K(z) = 1/\bar{z}$ and $s_L(z) = \bar{z} + 2ic$, so $s_L s_K(z) = 1/z + 2ic$. The fixed points solve

$$z^2 - 2icz - 1 = 0, \qquad z = ic \pm \sqrt{1 - c^2}.$$

- $c = 0.6$: $z = \pm 0.8 + 0.6i$, the real crossings.
- $c = 1$: $z = i$, a double root (tangency).
- $c = 1.7$: $z = 3.0748i$ and $z = 0.3252i$, the limiting points $(0, c \pm \sqrt{c^2 - 1})$, whose heights multiply to 1.

In the same case $c = 1.7$, the hidden crossings are $(\pm 1.3748i,\ 1.7)$ in $\mathbb{C}^2$. These also satisfy $K \star P \sim L \star P$; they are the two non-real members of Agree. A brute-force search of the real plane found no other agreement points. Over 2000 random real pairs (981 cutting, 1017 disjoint or nested) there were 0 mismatches with Theorem 7. Quilez's dots $(\pm\sqrt{c^2 - 1}, c)$ are the limiting points rotated by 90° about the foot $(0, c)$. This is the "i" of the hidden crossings made visible.

4 A note on $\mathbb{P}^3$ versus $\mathbb{P}^4$

Everything above lives in Möbius circle space, the real $\mathbb{P}^3$ with a quadric of signature $(3, 1)$. Its group is $PO(3, 1)$, which is the Möbius group of the sphere, generated by inversions [6]. Over $\mathbb{C}$, the quadric $\{Q = 0\} \subset \mathbb{CP}^3$ is $\mathbb{CP}^1 \times \mathbb{CP}^1$, and the point plane $\mathbb{C}^2$ sits inside it as an affine chart. This is where hidden crossings such as $(\pm 1.3748i, 1.7)$ live, so no extra dimension is needed. *Lie* sphere geometry instead puts *oriented* circles in $\mathbb{P}^4$, on the Lie quadric of signature $(3, 2)$, with group $O(3, 2)/\pm I$ [7]. One can define the same reflection kei there. However, the agreement theorem does not need orientation or $\mathbb{P}^4$, so we stay in $\mathbb{P}^3$.

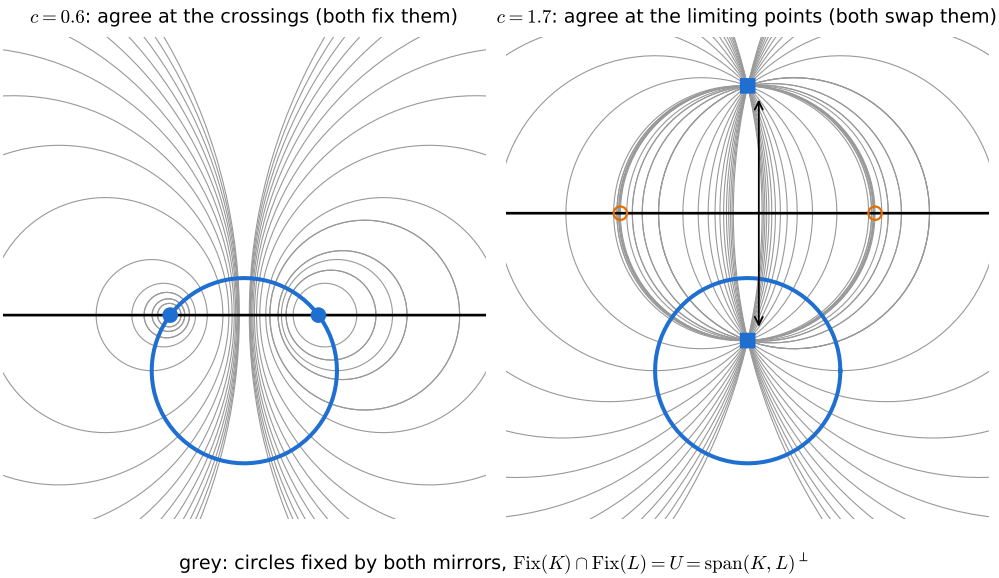


Figure 2: Agreement points (filled). Left: the line cuts, and the mirrors agree where they cross. Right: the line misses, the crossings are hidden (open circles mark Quilez's dots), and the mirrors agree at the limiting points, which both mirrors swap.

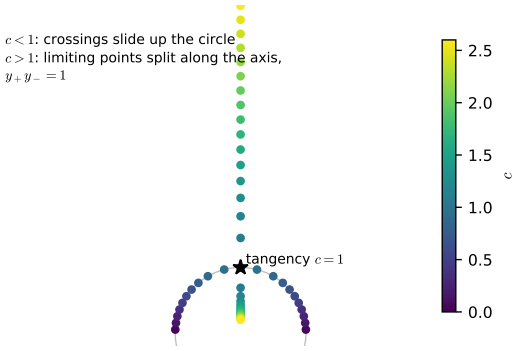


Figure 3: The two real agreement points as the line rises. There are always two, except at tangency.

References

[1] M. Takasaki, Abstractions of symmetric functions, *Tohoku Math. J.* 49 (1943) 143–207 (in Japanese).

[2] D. Joyce, A classifying invariant of knots, the knot quandle, *J. Pure Appl. Algebra* 23 (1982) 37–65, doi:10.1016/0022-4049(82)90077-9.

[3] O. Loos, Reflexion spaces and homogeneous symmetric spaces, *Bull. AMS* 73 (1967) 250–253.

[4] O. Loos, *Symmetric Spaces*, Benjamin, 1969.

[5] D. Stanovský, The origins of involutory quandles, arXiv:1506.02389.

[6] H. Schwerdtfeger, *Geometry of Complex Numbers*, Dover, 1979.

[7] T. E. Cecil, *Lie Sphere Geometry*, 2nd ed., Springer Universitext, 2008, doi:10.1007/978-0-387-74656-2.